

Qualification Pack



Hydroponics Technician

QP Code: AGR/Q0816

Version: 1.0

NSQF Level: 4

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AGR/Q0816: Hydroponics Technician

Brief Job Description

A Hydroponics Technician grows plants in a controlled environment using alternative materials instead of soil as per the hydroponics farming method. The individual is also responsible for setting and maintaining a hydroponic farm apart from maintaining, harvesting and marketing the plants grown at the farm.

Personal Attributes

The individual must be physically fit to work for long durations. The person must have attention to detail, problem-solving skills with the ability to work collaboratively with others to achieve the work objectives. The individual must know how to read and write.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [AGR/N0822: Operate and maintain the hydroponic system](#)
2. [AGR/N0857: Carry out crop management in hydroponics system](#)
3. [AGR/N0823: Carry out harvesting, post-harvest management and marketing activities](#)
4. [AGR/N9903: Maintain health and safety at the workplace](#)
5. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Agriculture
Sub-Sector	Agriculture Crop Production
Occupation	Landscaping, Gardening and Urban Farming
Country	India
NSQF Level	4
Credits	13
Aligned to NCO/ISCO/ISIC Code	NCO-2015/3142.9900

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Minimum Educational Qualification & Experience	12th grade Pass (Science) Or equivalent) OR 10th grade pass with 3 Years of experience relevant experience in in Agriculture & allied sector OR Previous relevant Qualification of NSQF Level (3.5) with 1.5 years of experience relevant experience in agri & allied sectors OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience relevant experience in agri & allied sectors
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	08/05/2028
NSQC Approval Date	08/05/2025
Version	1.0
Reference code on NQR	QG-04-AG-00289-2025-V2-ASCI
NQR Version	2

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AGR/N0822: Operate and maintain the hydroponic system

Description

This OS unit is about operating and maintaining the hydroponic system.

Scope

The scope covers the following :

- Select site for hydroponic farming
- Choose appropriate hydroponic system
- Maintain the hydroponic system

Elements and Performance Criteria

Select the site for hydroponic farming

To be competent, the user/individual on the job must be able to:

- PC1.** select a location with the required temperature and sunlight exposure, suitable to the selected plant/ crop variety
- PC2.** ensure the availability of various inputs such as water, electricity, fertilisers and labour for hydroponic farming
- PC3.** prepare the tabulation of the crop, following the recommended process

Choose the proper hydroponic system

To be competent, the user/individual on the job must be able to:

- PC4.** select the appropriate hydroponic technique for growing quality medium to obtain quality produce of the chosen crop
- PC5.** select an appropriate nutrient circulation method such as the Nutrient Film Technique, Deep Flow Technique (DFT), wick system, water culture, EBB and Flower etc
- PC6.** insert plants placed in plastic net pots, in the holes made in PVC pipes
- PC7.** use the recommended flooring material in the hydroponic farm, that is capable of soaking in any spillage and leakages
- PC8.** adopt the Nutrient Film Technique (NFT) for leafy plants for their faster growth (Selection of Hydroponics)
- PC9.** place plants in growing tubes and suspend them into water
- PC10.** maintain a reservoir to hold the nutrient solution and pump it as per the requirement to the tubes of growing plants as a back up
- PC11.** automate the nutrient flow system using the cyclic monitoring application, as required
- PC12.** ensure the availability of adequate lighting for the optimum growth of plants in the hydroponic system through the use of a relevant lighting system

Maintain the hydroponic system

To be competent, the user/individual on the job must be able to:

- PC13.** check the pipes for any salt deposits and remove (cleaning of NFT channels /pipes) it safely to prevent choking and ensure the nutrient solution flows without obstruction through the pipes

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- PC14.** check and ensure that the roots of plants get an appropriate quantity of nutrient solution for healthy growth
- PC15.** clean and monitor the condition of motors controlling water tanks, pumps, and growing trays
- PC16.** use clean water in the hydroponic system and maintain the pH level of the water by using the recommended treatment
- PC17.** clean the floor and maintain it dry in the hydroponic system
- PC18.** use mesh filters for regular cleaning
- PC19.** check the pH & EC regularly and follow the recommended practices to adjust it to the required level in the hydroponic system and nutrient solution, as required
- PC20.** use the oxygen meter with mobile application support and alarm to monitor the level of dissolved oxygen in the nutrient solution and follow the recommended practices to maintain or adjust it
- PC21.** clean the hydroponic system regularly using agriculture grade acid or other recommended treatment, and flush the system with clean water before replanting
- PC22.** use auto-cleaning systems for cleaning in large-scale hydroponic systems to save time and achieve better productivity
- PC23.** carry out artificial pollination using blowers, and use mechanical vibrators to improve air quality within the protected hydroponic system structures
- PC24.** replace damaged parts or components as needed

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** importance of selecting a site with suitable temperature and sunlight exposure for the chosen crop.
- KU2.** effect of environmental conditions (humidity, airflow, shading) on plant growth
- KU3.** applicable work modalities and requirements for hydroponic farming
- KU4.** assessing the availability of water, electricity, fertilizers, and labor
- KU5.** water quality parameters (pH, electrical conductivity, dissolved oxygen, etc.).
- KU6.** following the recommended process for structuring the hydroponic farm layout
- KU7.** matching circulation methods with crop type and system size
- KU8.** importance of automation in nutrient flow regulation
- KU9.** different types of hydroponics systems
- KU10.** process of hydroponic cultivation using the relevant automated systems
- KU11.** the application of Deep Flow Technique (DFT) and Nutrient Film Technique (NFT)
- KU12.** different types of nutrient solutions, their composition and application
- KU13.** the impact of different climatic conditions on hydroponic farming
- KU14.** use of hydroponics kits and home hydroponic unit
- KU15.** the indoor hydroponic system and relevant plants
- KU16.** the criteria for selecting the location, plant varieties and substrate/ growth medium for hydroponic farming
- KU17.** ensuring adequate artificial lighting (LEDs, UV lamps) for optimal plant growth

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- KU18.** oxygen and nutrient management
- KU19.** system sanitization and cleaning
- KU20.** system repairs and component replacement

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to learn about the latest developments in the field of work
- GS3.** communicate politely and professionally
- GS4.** take quick decisions to resolve work-related issues and minimise the impact on productivity
- GS5.** listen attentively to understand the information being shared
- GS6.** plan and prioritise tasks to ensure timely completion
- GS7.** evaluate all possible solutions to a problem to select the best one
- GS8.** co-ordinate with the co-workers to achieve the work objectives

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Select the site for hydroponic farming</i>	5	5	-	8
PC1. select a location with the required temperature and sunlight exposure, suitable to the selected plant/ crop variety	-	-	-	-
PC2. ensure the availability of various inputs such as water, electricity, fertilisers and labour for hydroponic farming	-	-	-	-
PC3. prepare the tabulation of the crop, following the recommended process	-	-	-	-
<i>Choose the proper hydroponic system</i>	10	15	-	10
PC4. select the appropriate hydroponic technique for growing quality medium to obtain quality produce of the chosen crop	-	-	-	-
PC5. select an appropriate nutrient circulation method such as the Nutrient Film Technique, Deep Flow Technique (DFT), wick system, water culture, EBB and Flower etc	-	-	-	-
PC6. insert plants placed in plastic net pots, in the holes made in PVC pipes	-	-	-	-
PC7. use the recommended flooring material in the hydroponic farm, that is capable of soaking in any spillage and leakages	-	-	-	-
PC8. adopt the Nutrient Film Technique (NFT) for leafy plants for their faster growth (Selection of Hydroponics)	-	-	-	-
PC9. place plants in growing tubes and suspend them into water	-	-	-	-
PC10. maintain a reservoir to hold the nutrient solution and pump it as per the requirement to the tubes of growing plants as a back up	-	-	-	-
PC11. automate the nutrient flow system using the cyclic monitoring application, as required	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. ensure the availability of adequate lighting for the optimum growth of plants in the hydroponic system through the use of a relevant lighting system	-	-	-	-
<i>Maintain the hydroponic system</i>	15	20	-	12
PC13. check the pipes for any salt deposits and remove (cleaning of NFT channels /pipes) it safely to prevent choking and ensure the nutrient solution flows without obstruction through the pipes	-	-	-	-
PC14. check and ensure that the roots of plants get an appropriate quantity of nutrient solution for healthy growth	-	-	-	-
PC15. clean and monitor the condition of motors controlling water tanks, pumps, and growing trays	-	-	-	-
PC16. use clean water in the hydroponic system and maintain the pH level of the water by using the recommended treatment	-	-	-	-
PC17. clean the floor and maintain it dry in the hydroponic system	-	-	-	-
PC18. use mesh filters for regular cleaning	-	-	-	-
PC19. check the pH & EC regularly and follow the recommended practices to adjust it to the required level in the hydroponic system and nutrient solution, as required	-	-	-	-
PC20. use the oxygen meter with mobile application support and alarm to monitor the level of dissolved oxygen in the nutrient solution and follow the recommended practices to maintain or adjust it	-	-	-	-
PC21. clean the hydroponic system regularly using agriculture grade acid or other recommended treatment, and flush the system with clean water before replanting	-	-	-	-
PC22. use auto-cleaning systems for cleaning in large-scale hydroponic systems to save time and achieve better productivity	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC23. carry out artificial pollination using blowers, and use mechanical vibrators to improve air quality within the protected hydroponic system structures	-	-	-	-
PC24. replace damaged parts or components as needed	-	-	-	-
NOS Total	30	40	-	30

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National Occupational Standards (NOS) Parameters

NOS Code	AGR/N0822
NOS Name	Operate and maintain the hydroponic system
Sector	Agriculture
Sub-Sector	Agriculture Crop Production
Occupation	Landscaping, Gardening and Urban Farming
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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AGR/N0857: Carry out crop management in hydroponics system

Description

This OS unit is about preparing the medium, seedlings and carrying out irrigation

Scope

The scope covers the following :

- Prepare the growing medium
- Prepare seedlings for hydroponic farming
- Carry out irrigation and fertigation
- Monitor pests & diseases

Elements and Performance Criteria

Prepare the growing medium

To be competent, the user/individual on the job must be able to:

- PC1.** select the appropriate growing medium, except soil for seed germination, ensuring the medium has the required characteristics such as moderate fertility, water holding capacity, good aeration capacity, etc.
- PC2.** use coco peat, rice husk to raise seedlings, avoiding the use of any medium to grow agronomical crops such as wheat, paddy, maize, barley, oat, etc. for nursery and fodder purposes
- PC3.** sterilize the growing medium before use, according to the applicable sterilization limit

Prepare seedlings for hydroponic farming

To be competent, the user/individual on the job must be able to:

- PC4.** select nursery containers for the growth of plants such as clay pots, plastic pots, or trays, as required
- PC5.** clean and sterilize the pots and trays before sowing seeds
- PC6.** choose appropriate crop according to the system
- PC7.** position the seeds appropriately on growing trays to prevent waterlogging in them
- PC8.** soak good quality seeds as per the requirement and use an appropriate time-management technique to avoid over-soaking of seeds
- PC9.** apply the nutrient solution in the recommended quantity to the trays and pots and sow the seeds
- PC10.** use pest and disease-free seedlings, and planting material for the establishment of hydroponic crops
- PC11.** maintain seedlings in the trays and pots for the recommended duration, protecting them from pests and disease and ensuring effective nutrient management
- PC12.** transplant the seedlings at the appropriate stage of their growth to be transplanted in the appropriate hydroponic system

Carry out irrigation and fertigation

To be competent, the user/individual on the job must be able to:

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- PC13.** determine the amount of fertilizers to be used for fertigation based on the selected crop, its stages of growth, and the type of selected hydroponics technique
- PC14.** select the relevant fertigation technique for use, as appropriate
- PC15.** mix the appropriate fertilizers with the daily water requirement or by using a fertigation system/nutrient tank, according to the selected fertigation technique
- PC16.** follow fertigation schedule as per group growth stage
- PC17.** maintain the record of irrigation and fertigation of crop
- PC18.** ensure the required macro and micronutrients are available to plants and crops according to different stages of their growth

Monitor pests & diseases

To be competent, the user/individual on the job must be able to:

- PC19.** monitor the crops for any signs of pests and diseases and take appropriate remedial measures
- PC20.** implement integrated pest management (IPM) strategies.
- PC21.** use organic or chemical treatments as required

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** different types of growing mediums (coco peat, perlite, rock wool, rice husk, etc.).
- KU2.** properties of an ideal medium: moderate fertility, water retention, aeration
- KU3.** mediums unsuitable for hydroponics (e.g., soil, wheat, paddy, maize, barley, oat).
- KU4.** the techniques for raising seedlings using coco peat and rice husk
- KU5.** the selection of nursery containers (clay pots, plastic pots, trays) based on crop type.
- KU6.** the Importance of sterilizing growing mediums and containers before use
- KU7.** proper positioning of seeds in growing trays to prevent waterlogging
- KU8.** soaking methods for high-quality seeds to avoid over-soaking issues.
- KU9.** nutrient solution application for seedling trays and pots.
- KU10.** identifying healthy, pest-free seedlings and planting material.
- KU11.** managing pests and diseases in early-stage seedlings.
- KU12.** transplanting seedlings at the correct growth stage into hydroponic systems
- KU13.** determining fertilizer quantity based on crop type, growth stage, and hydroponic technique.
- KU14.** developing fertigation schedules as per crop growth stages
- KU15.** identifying common hydroponic crop pests (aphids, whiteflies, mites, etc.)
- KU16.** integrated Pest Management (IPM) Strategies
- KU17.** the essential plant nutrient elements and the criteria for essentiality
- KU18.** the symptoms of plant nutrient element deficiency and excess
- KU19.** the frequency and rate of nutrient solution dosing of plant roots
- KU20.** characteristics of high-quality irrigation water
- KU21.** process of filtering and sterilizing the water and nutrient solution
- KU22.** process of raising seedlings for hydroponics farming

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KU23. appropriate pH & EC levels to be maintained in the hydroponics system

KU24. process to be followed for the production of contamination-free crop

KU25. process of application of organic/chemical pest treatments

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. maintain work-related notes and records

GS2. read the relevant literature to learn about the latest developments in the field of work

GS3. communicate politely and professionally

GS4. take quick decisions to resolve work-related issues and minimise the impact on productivity

GS5. listen attentively to understand the information being shared

GS6. plan and prioritise tasks to ensure timely completion

GS7. evaluate all possible solutions to a problem to select the best one

GS8. co-ordinate with the co-workers to achieve the work objectives

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare the growing medium</i>	5	10	-	5
PC1. select the appropriate growing medium, except soil for seed germination, ensuring the medium has the required characteristics such as moderate fertility, water holding capacity, good aeration capacity, etc.	-	-	-	-
PC2. use coco peat, rice husk to raise seedlings, avoiding the use of any medium to grow agronomical crops such as wheat, paddy, maize, barley, oat, etc. for nursery and fodder purposes	-	-	-	-
PC3. sterilize the growing medium before use, according to the applicable sterilization limit	-	-	-	-
<i>Prepare seedlings for hydroponic farming</i>	10	20	-	10
PC4. select nursery containers for the growth of plants such as clay pots, plastic pots, or trays, as required	-	-	-	-
PC5. clean and sterilize the pots and trays before sowing seeds	-	-	-	-
PC6. choose appropriate crop according to the system	-	-	-	-
PC7. position the seeds appropriately on growing trays to prevent waterlogging in them	-	-	-	-
PC8. soak good quality seeds as per the requirement and use an appropriate time-management technique to avoid over-soaking of seeds	-	-	-	-
PC9. apply the nutrient solution in the recommended quantity to the trays and pots and sow the seeds	-	-	-	-
PC10. use pest and disease-free seedlings, and planting material for the establishment of hydroponic crops	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. maintain seedlings in the trays and pots for the recommended duration, protecting them from pests and disease and ensuring effective nutrient management	-	-	-	-
PC12. transplant the seedlings at the appropriate stage of their growth to be transplanted in the appropriate hydroponic system	-	-	-	-
<i>Carry out irrigation and fertigation</i>	8	5	-	8
PC13. determine the amount of fertilizers to be used for fertigation based on the selected crop, its stages of growth, and the type of selected hydroponics technique	-	-	-	-
PC14. select the relevant fertigation technique for use, as appropriate	-	-	-	-
PC15. mix the appropriate fertilizers with the daily water requirement or by using a fertigation system/nutrient tank, according to the selected fertigation technique	-	-	-	-
PC16. follow fertigation schedule as per group growth stage	-	-	-	-
PC17. maintain the record of irrigation and fertigation of crop	-	-	-	-
PC18. ensure the required macro and micronutrients are available to plants and crops according to different stages of their growth	-	-	-	-
<i>Monitor pests & diseases</i>	7	5	-	7
PC19. monitor the crops for any signs of pests and diseases and take appropriate remedial measures	-	-	-	-
PC20. implement integrated pest management (IPM) strategies.	-	-	-	-
PC21. use organic or chemical treatments as required	-	-	-	-
NOS Total	30	40	-	30

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National Occupational Standards (NOS) Parameters

NOS Code	AGR/N0857
NOS Name	Carry out crop management in hydroponics system
Sector	Agriculture
Sub-Sector	
Occupation	Landscaping, Gardening and Urban Farming
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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AGR/N0823: Carry out harvesting, post-harvest management and marketing activities

Description

This OS unit is about carrying out harvesting, post-harvest management and marketing of produce grown in hydroponic farming.

Scope

The scope covers the following :

- Carry out harvesting activities
- Perform post-harvest management
- Market the produce

Elements and Performance Criteria

Carry out harvesting activities

To be competent, the user/individual on the job must be able to:

- PC1.** check the maturity of plants and crops for harvesting
- PC2.** carry out harvesting using the relevant tools and implements, ensuring minimum damage to crops/plants during harvesting
- PC3.** discard any damaged or disfigured plants
- PC4.** carry out sorting and grading on the basis of applicable parameters such as colour, appearance, ripeness, etc.
- PC5.** tag the harvested plants/ crop for identification
- PC6.** tag the harvested plants/crop for identification
- PC7.** maintain the record of harvesting schedule and period of cultivation of crops/plants

Perform post-harvest management

To be competent, the user/individual on the job must be able to:

- PC8.** store the harvested plants/crops in a dry storage area with the recommended temperature and humidity, ensuring hygiene and aeration
- PC9.** use humidity monitors to monitor the humidity

Market the produce

To be competent, the user/individual on the job must be able to:

- PC10.** track and analyse the information related to the wholesale and retail price of the produce
- PC11.** identify the market and buyers of the produce such as e-trading platforms, cooperatives, local traders, exporters, etc.
- PC12.** coordinate and negotiate with the potential buyers to secure a profitable price for the produce
- PC13.** pack the produce in the appropriate packing material and label it with the relevant information

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- PC14.** arrange an appropriate mode of transport for safe and hygienic delivery of produce, ensuring optimum aeration during transit
- PC15.** coordinate with the transporter for safe and timely delivery to the buyer
- PC16.** process the payment using the buyer-preferred e-payment method
- PC17.** maintain the manual and/ or electronic record of sales and payments using the physical registers and/ or the relevant computer system
- PC18.** ensure compliance with the applicable regulations in the marketing of produce

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the relevant government schemes with the provision of subsidies for the promotion and marketing of hydroponic farming
- KU2.** the relevant state units and other financial institutions to be approached to avail the relevant government schemes
- KU3.** the appropriate marketing channels for selling the produce from hydroponic farms, and their constraints
- KU4.** the applicable procedures used for harvesting, storage, and logistics
- KU5.** the applicable harvesting schedules and techniques
- KU6.** the relevant aeration techniques
- KU7.** the applicable packaging techniques and labelling requirements
- KU8.** the signs of maturity in different types of crops and plants
- KU9.** use of the relevant tools, equipment and accessories for harvesting and collecting the produce
- KU10.** the recommended practices to protect the produce from damage and contamination
- KU11.** relevant parameters to sort and grade the harvested produce on
- KU12.** the importance of storing the harvested produce under the recommended temperature, humidity and hygienic conditions
- KU13.** how to identify, connect and negotiate with potential buyers
- KU14.** appropriate material to pack the produce and the relevant labelling requirements
- KU15.** use of different types of e-payment methods
- KU16.** the importance of maintaining the record of sales and payments
- KU17.** how to maintain manual and electronic records using the physical registers and the relevant computer application respectively
- KU18.** basic accounting practices such as calculating the expenditure and cost of production, and benefit-cost (B:C) ratio, etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records

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- GS2.** read the relevant literature to get the latest updates about the field of work
- GS3.** listen attentively to understand the information being shared
- GS4.** communicate clearly and politely
- GS5.** plan and prioritise tasks for effective time management
- GS6.** take quick decisions to deal with any emergencies/ accidents or disruptions to work
- GS7.** co-ordinate with co-workers to achieve work objectives
- GS8.** evaluate all possible solutions to a problem to select the best one

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Carry out harvesting activities</i>	12	14	-	12
PC1. check the maturity of plants and crops for harvesting	-	-	-	-
PC2. carry out harvesting using the relevant tools and implements, ensuring minimum damage to crops/plants during harvesting	-	-	-	-
PC3. discard any damaged or disfigured plants	-	-	-	-
PC4. carry out sorting and grading on the basis of applicable parameters such as colour, appearance, ripeness, etc.	-	-	-	-
PC5. tag the harvested plants/ crop for identification	-	-	-	-
PC6. tag the harvested plants/crop for identification	-	-	-	-
PC7. maintain the record of harvesting schedule and period of cultivation of crops/plants	-	-	-	-
<i>Perform post-harvest management</i>	6	8	-	4
PC8. store the harvested plants/crops in a dry storage area with the recommended temperature and humidity, ensuring hygiene and aeration	-	-	-	-
PC9. use humidity monitors to monitor the humidity	-	-	-	-
<i>Market the produce</i>	12	18	-	14
PC10. track and analyse the information related to the wholesale and retail price of the produce	-	-	-	-
PC11. identify the market and buyers of the produce such as e-trading platforms, cooperatives, local traders, exporters, etc.	-	-	-	-
PC12. coordinate and negotiate with the potential buyers to secure a profitable price for the produce	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. pack the produce in the appropriate packing material and label it with the relevant information	-	-	-	-
PC14. arrange an appropriate mode of transport for safe and hygienic delivery of produce, ensuring optimum aeration during transit	-	-	-	-
PC15. coordinate with the transporter for safe and timely delivery to the buyer	-	-	-	-
PC16. process the payment using the buyer-preferred e-payment method	-	-	-	-
PC17. maintain the manual and/ or electronic record of sales and payments using the physical registers and/ or the relevant computer system	-	-	-	-
PC18. ensure compliance with the applicable regulations in the marketing of produce	-	-	-	-
NOS Total	30	40	-	30

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National Occupational Standards (NOS) Parameters

NOS Code	AGR/N0823
NOS Name	Carry out harvesting, post-harvest management and marketing activities
Sector	Agriculture
Sub-Sector	Agriculture Crop Production
Occupation	Landscaping, Gardening and Urban Farming
NSQF Level	4
Credits	2
Version	3.0
Last Reviewed Date	08/05/2025
Next Review Date	08/05/2028
NSQC Clearance Date	08/05/2025

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AGR/N9903: Maintain health and safety at the workplace

Description

This OS is about maintaining health and safety of self and other co-workers at the workplace

Scope

The scope covers the following :

- Maintain personal hygiene
- Maintain clean and safe workplace
- Administer appropriate emergency procedures

Elements and Performance Criteria

Maintain personal hygiene

To be competent, the user/individual on the job must be able to:

- PC1.** wash hands, legs and face with soap/alcohol based sanitizer at reasonable intervals
- PC2.** wash the worn clothes with soap and sun dry before use next time
- PC3.** ensure the face is covered with mask or three layers of cloth-piece
- PC4.** follow the workplace sanitization norms including distancing from sick people

Maintain clean and safe workplace

To be competent, the user/individual on the job must be able to:

- PC5.** carry out basic safety checks before operation of all tools, implements, and machinery and report identified hazards to the supervisor
- PC6.** wear appropriate Personal Protective Equipment (PPE) while performing work in accordance with the workplace policy
- PC7.** follow the instructions mentioned on the labels of chemicals/pesticides/fumigants etc to avoid hazards
- PC8.** assess risks prior to performing manual handling jobs, and work according to currently recommended safe practices
- PC9.** sanitize equipment, tools and machinery before and after use
- PC10.** use equipment and materials safely and correctly and return the same to designated storage after use
- PC11.** dispose waste safely and correctly in the designated area
- PC12.** recognize risks to bystanders and take required action to reduce the risks
- PC13.** work in a manner which minimizes environmental damage, ensuring all procedures and instructions for controlling risks are followed
- PC14.** report any accidents, incidents or problems without delay to an appropriate person and take necessary immediate action to reduce further danger
- PC15.** follow government / workplace advisories incase of outbreak of any disease/disaster

Administer appropriate emergency procedures

To be competent, the user/individual on the job must be able to:

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- PC16.** follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to the location of emergency, as per the workplace requirements
- PC17.** use emergency equipment in accordance with manufacturer's specifications and workplace requirements
- PC18.** provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques
- PC19.** recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate
- PC20.** report details of first aid administered in accordance with workplace procedures

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** relevant legislation, standards, policies, and procedures at work
- KU2.** relevant health and safety requirements applicable to the work environment
- KU3.** own job role and responsibilities and sources of information pertaining to work
- KU4.** who to approach for support in order to obtain work related information, clarifications and support
- KU5.** importance of following health, hygiene, safety and quality standards and the impact of not following the standards on consumers and the business
- KU6.** personal hygiene and fitness requirement
- KU7.** importance of sanitization of the workplace
- KU8.** types of Personal Protective Equipment (PPE) required at the workplace and their importance
- KU9.** the correct and safe way to use materials and equipment required for the work
- KU10.** the importance of good housekeeping at the workplace
- KU11.** safe waste disposal methods
- KU12.** methods for minimizing environmental damage during work
- KU13.** the risks to health and safety including contagious diseases and the measures to be taken to control those risks in the area of work
- KU14.** workplace procedures and requirements for the prevention and treatment of workplace injuries/illnesses.
- KU15.** basic emergency first aid procedure
- KU16.** local emergency services
- KU17.** why accidents, incidents and problems should be reported and the appropriate actions to be taken

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** record the data as per the requirement
- GS2.** report problems to the appropriate personnel in a timely manner

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- GS3.** read instruction manual for hand tool and equipments
- GS4.** communicate clearly and effectively with co-workers, and other stakeholders
- GS5.** comprehend information shared by senior people and experts
- GS6.** make decisions pertaining to personal hygiene and safety
- GS7.** schedule daily activities and draw up priorities
- GS8.** manage relationships with co-workers, manager and other stakeholders
- GS9.** assess situation and identify appropriate control measures

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Maintain personal hygiene</i>	10	5	-	10
PC1. wash hands, legs and face with soap/alcohol based sanitizer at reasonable intervals	-	-	-	-
PC2. wash the worn clothes with soap and sun dry before use next time	-	-	-	-
PC3. ensure the face is covered with mask or three layers of cloth-piece	-	-	-	-
PC4. follow the workplace sanitization norms including distancing from sick people	-	-	-	-
<i>Maintain clean and safe workplace</i>	15	15	-	15
PC5. carry out basic safety checks before operation of all tools, implements, and machinery and report identified hazards to the supervisor	-	-	-	-
PC6. wear appropriate Personal Protective Equipment (PPE) while performing work in accordance with the workplace policy	-	-	-	-
PC7. follow the instructions mentioned on the labels of chemicals/pesticides/fumigants etc to avoid hazards	-	-	-	-
PC8. assess risks prior to performing manual handling jobs, and work according to currently recommended safe practices	-	-	-	-
PC9. sanitize equipment, tools and machinery before and after use	-	-	-	-
PC10. use equipment and materials safely and correctly and return the same to designated storage after use	-	-	-	-
PC11. dispose waste safely and correctly in the designated area	-	-	-	-
PC12. recognize risks to bystanders and take required action to reduce the risks	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. work in a manner which minimizes environmental damage, ensuring all procedures and instructions for controlling risks are followed	-	-	-	-
PC14. report any accidents, incidents or problems without delay to an appropriate person and take necessary immediate action to reduce further danger	-	-	-	-
PC15. follow government / workplace advisories incase of outbreak of any disease/disaster	-	-	-	-
<i>Administer appropriate emergency procedures</i>	15	5	-	10
PC16. follow procedures for dealing with accidents, fires and emergencies, including communicating location and directions to the location of emergency, as per the workplace requirements	-	-	-	-
PC17. use emergency equipment in accordance with manufacturer's specifications and workplace requirements	-	-	-	-
PC18. provide treatment appropriate to the patient's injuries in accordance with recognized first aid techniques	-	-	-	-
PC19. recover (if practical), clean, inspect/test, refurbish, replace and store the first aid equipment as appropriate	-	-	-	-
PC20. report details of first aid administered in accordance with workplace procedures	-	-	-	-
NOS Total	40	25	-	35

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	AGR/N9903
NOS Name	Maintain health and safety at the workplace
Sector	Agriculture
Sub-Sector	Generic
Occupation	Generic
NSQF Level	4
Credits	1
Version	4.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down the proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on the knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/optional set of NOS.
4. Individual assessment agencies will create unique question papers for the theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score a minimum of 70% of % aggregate marks to successfully clear the assessment.

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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
AGR/N0822. Operate and maintain the hydroponic system	30	40	-	30	100	35
AGR/N0857. Carry out crop management in hydroponics system	30	40	-	30	100	30
AGR/N0823. Carry out harvesting, post-harvest management and marketing activities	30	40	-	30	100	25
AGR/N9903. Maintain health and safety at the workplace	40	25	-	35	100	5
DGT/VSQ/N0102. Employability Skills (60 Hours)	20	30	-	-	50	5
Total	150	175	-	125	450	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
PPE	Personal Protective Equipment
PPE	Personal Protective Equipment

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.